Exploring Population Data and Long-Term Trends

In this assignment, you will use a long-term dataset on female winter flounder to investigate changes in age composition, water temperature, and catch per unit effort (CPUE). You will organize data, create visualizations, fit simple linear regression models, examine residuals, and interpret the biological meaning and limitations of the results.

Learning Objectives

After completing this assignment, you should be able to

- import and organize data from an Excel workbook;
- compute annual age-class percentages;
- create clear visualizations of population age structure;
- fit and interpret simple linear regression models;
- compute and examine residuals;
- assess residual normality using the Shapiro–Wilk test;
- relate environmental variables to biological observations;
- communicate statistical results in biological context.

1 Import and Inspect the Data

Begin by importing the required packages.

```
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import linregress, shapiro
```

Read the worksheet containing annual counts of females by age class.

```
input_file = "data.xlsx"

df = pd.read_excel(
    input_file,
    sheet_name="AllFemales",
    index_col=0
)

# Inspect the first several rows
df.head()
```

Complete the following tasks.

1. Record the number of years and age classes in the dataset.
2. Identify missing values if any.
3. Explain what the rows and columns represent.

2 Annual Age Composition

For each year, convert the observed counts to percentages.

If $N_{i,t}$ denotes the number of females in age class i during year t, define

$$P_{i,t} = \frac{N_{i,t}}{\sum_j N_{j,t}} \times 100.$$

Complete the code below.

```
summary_table = df.copy()

# TODO:
# Compute the percentage represented by each age class
# in each year.

age_percentages =

# Round to two decimal places
age_percentages =
```

Check that the percentages sum to approximately 100 for every year.

```
# TODO:
# Verify row sums
```

Answer the following questions.

1. Which age classes are most common?
2. Does the age composition appear stable through time?
3. Are there any years that appear unusual?
4. Why may percentages be more appropriate than raw counts for comparing age composition among years?

3 Visualizing Annual Age Structure

Create a stacked bar plot showing annual age composition.

Starter code is provided below.

```
ax = age_percentages.plot(
    kind='bar',
    stacked=True,
    figsize=(12, 7),
```

```
    colormap='tab20'
)

plt.xlabel('Year', fontsize=16)
plt.ylabel('Percentage (%)', fontsize=16)
plt.xticks(rotation=45, fontsize=14)
plt.yticks(fontsize=14)

plt.legend(
    title='Age Group',
    title_fontsize=14,
    fontsize=14,
    bbox_to_anchor=(1.05, 1),
    loc='upper left'
)

plt.tight_layout()
plt.show()
```

You may also want to create an alternative line plot with one line for each age class. Your line plot will use year on the horizontal axis, use percentage on the vertical axis, include a readable legend, and use markers or line styles that distinguish age classes.

4 Trend Analysis by Age Class

For each age class, fit the simple linear regression model

$$P_i(t) = a_i + b_i t + \varepsilon_i(t),$$

where $P_i(t)$ is the percentage of females in age class i during year t.

Use the following scaffold.

```
results = []

for age_group in age_order:
    x = age_percentages.index
    y = age_percentages[age_group]

    # TODO:
    # Fit a simple linear regression using linregress.
    slope, intercept, r_value, p_value, std_err =

    # TODO:
    # Compute fitted values.
    predicted =

    # TODO:
    # Compute residuals.
    residuals =

    r_squared =

    # TODO:
```

```
    # Apply the Shapiro-Wilk test to residuals.
    shapiro_stat, shapiro_p =

    results.append({
        "Age group": age_group,
        "Slope": slope,
        "Intercept": intercept,
        "R_squared": r_squared,
        "p_value": p_value,
        "Shapiro_p": shapiro_p
    })
```

For each age class, create a figure containing

1. the observed percentages and fitted regression line;
2. the regression residuals.

A possible plotting scaffold is

```
fig, (ax_main, ax_resid) = plt.subplots(
    2, 1,
    figsize=(10, 8),
    gridspec_kw={'height_ratios': [2, 1]}
)

ax_main.scatter(x, y, s=50, label=f'Age {age_group} (%)')
ax_main.plot(
    x,
    predicted,
    linestyle='--',
    linewidth=2.5,
    label='Linear fit'
)

ax_main.set_ylabel('Percentage (%)', fontsize=16)
ax_main.tick_params(axis='both', labelsize=14)
ax_main.legend(fontsize=14)
ax_main.grid(True)

ax_resid.bar(x, residuals)
ax_resid.axhline(0, linestyle='--', linewidth=1.5)
ax_resid.set_xlabel('Year', fontsize=16)
ax_resid.set_ylabel('Residuals', fontsize=16)
ax_resid.tick_params(axis='both', labelsize=14)
ax_resid.grid(True)

plt.tight_layout()

# Save as a vector graphic
fig.savefig(
    f"age_{age_group}_regression.pdf",
    bbox_inches="tight"
)
```

```
plt.show()
```

Create a summary table from the list `results`.

```
results_df = pd.DataFrame(results)
results_df
```

Interpret the results.

1. Which age classes have positive estimated slopes?
2. Which age classes have negative estimated slopes?
3. Which trends are statistically significant at the 5% level?

5 Water Temperature Trend

Read the worksheet containing annual water temperature.

```
temperature_df = pd.read_excel(
    input_file,
    sheet_name="Temperature",
    index_col=0
)

temperature_df.head()
```

Identify the temperature column and fit

$$T(t) = a + bt + \varepsilon(t).$$

Complete the following tasks.

1. Plot annual water temperature against year.
2. Add the fitted regression line.
3. Report the slope, R^2, and p-value.
4. Plot and examine the residuals.
5. Apply a Shapiro–Wilk test to the residuals.

Starter code:

```
x = temperature_df.index
y = temperature_df.iloc[:, 0]

# TODO:
slope, intercept, r_value, p_value, std_err =

# TODO:
predicted =

# TODO:
residuals =
```

Discuss whether the data provide evidence of a long-term temperature trend.

6 Female CPUE and Water Temperature

Read the worksheet containing tow information.

```
tow_df = pd.read_excel(
    input_file,
    sheet_name="TowCount",
    index_col=0
)

tow_df.head()
```

Compute female catch per unit effort using the definition

$$\text{CPUE} = \frac{\text{female count}}{\text{number of tows}}.$$

Then merge the CPUE and temperature data by year.

```
# TODO:
# Compute female CPUE.

# TODO:
# Merge CPUE and temperature data using their common years.
```

Fit the regression model

$$\text{CPUE} = a + b(\text{Temperature}) + \varepsilon.$$

Produce a scatter plot with a fitted line and report slope, intercept, R^2, p-value, residual normality test.

Answer the following questions.

1. What is the direction of the estimated relationship?
2. Is the relationship statistically significant?
3. Does the analysis establish causation? Explain.
4. What limitations arise from using annual averages?

7 Exporting Results

Export the annual counts and percentages to an Excel workbook.

```
combined = pd.concat(
    [summary_table, age_percentages],
    axis=1,
    keys=['Count', 'Percentage']
)

combined = combined.swaplevel(axis=1).sort_index(
    axis=1,
    level=0
)
```

```
with pd.ExcelWriter(
    "female_age_summary.xlsx",
    engine="openpyxl"
) as writer:
    summary_table.to_excel(writer, sheet_name="Counts")
    age_percentages.to_excel(writer, sheet_name="Percentages")
    combined.to_excel(writer, sheet_name="Combined")
```

Open the exported workbook and verify that the worksheets and values are correct.

Interpretation

Write a short report addressing the following questions.

1. Are there any temporal patterns in the age composition?
2. Is there evidence of a long-term trend in water temperature?
3. Is there evidence of a relationship between temperature and CPUE?
4. Which conclusions are directly supported by the data?
5. Which conclusions would require additional data or a more sophisticated modeling approach?

Deliverables

1. A completed Jupyter notebook;
2. An exported Excel summary workbook;
3. All required figures;
4. A one- to two-page report interpreting the results.